Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/29/2014
PI Name:	Roux, Stanley Ph.D.		
Project Title:	Rapid Signaling Changes Induced by Gravity in Cells of the Fern Ceratopteris richardii		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	78712-1240	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Greg Ph.D. (University of Texas)		
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Task Description:	Single-celled spores of the fern Ceratopteris richardii respond within seconds to both the direction and magnitude of gravity stimuli by altering the direction and magnitude of a trans-cell Ca2+ current that is required for the gravity response. This proposal aims to clarify the molecular mechanisms by which gravity alters the trans-cell current, and identify key signaling steps that link the current to cellular polarity changes directed by gravity. This focus on gravity signaling addresses an overarching question in the Space Biology Science Plan. Ca2+-uptake channels and Ca2+-ATPase efflux pumps drive the Ca2+ current across spore cells. Transcriptomic and proteomic analyses of genes and proteins highly expressed in spores at the peak of the Ca2+ current have identified likely candidates for the pumps and channels. Immunological approaches and yeast-based activity assays will be used to determine whether these transporters are localized on the plasma membrane and are regulated post-translationally. Recent evidence indicates that extracellular eATP [eATP], a newly discovered signaling agent, helps control the gravity response in Ceratopteris, just		

	as it helps regulate gravitropism in Arabidopsis roots. Because eATP controls the activity of Ca2+ channels in plants, we will test the regulatory link between eATP and the activity of Ca2+ channels that is crucial for the gravity response in Ceratopteris. Ca2+-binding proteins such as calmodulins, annexins, and Ca2+-dependent protein kinases are highly expressed in Ceratopteris at the peak of the Ca2+ current. They are the most likely signal transducers that help mediate the downstream cell polarization events guided by gravity, and their regulatory role in the gravity response will be tested. Expected results will generate hypotheses that predict molecular responses likely to occur in Ceratopteris cells in response to the micro-g environment, and the proposal will describe ISS experiments that could test these predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The germinating spore of Ceratopteris is an especially valuable model system for studying the cellular bases of gravity-induced graviresponses on Earth, because both the sensing of gravity and the response occur in the same cell. On the cellular level mechanisms of sensory responses are very similar in plant and animal cells, thus the bottom-to-top Ca2+ current that develops early during spore germination as one of the first polarized events induced by gravity in the cell may be an early event also in animal cells known to respond to gravity, such as muscle and bone cells.. Our investigation into candidate molecules likely to be associated with both the efflux and influx phases of this current will help clarify how this current is controlled. A number of studies have documented the importance of Ca2+ currents for polarized growth in both animal and plant cells; however, there are few prior studies that have identified and analyzed the molecular components of these currents. Moreover, while it is generally accepted that Ca2+ is a second messenger that participates in a myriad of plant and animal sensory responses, there is still little known about the molecular identity of the Ca2+ channels that mediate many of these responses, especially responses to gravitational stimuli. The proposed tests of Ca2+ pump and channel function and regulation, including possibly channel regulation by eATP, will not only provide novel insights into how the gravity-directed Ca2+ current originates, but also further advance an understanding of channel structures. The results will set the stage for future assays of post-translational changes that occur during space flight in the Ca2+-ATPase pumps and/or mechanosensitive channels in fern and possibly also in some animal cells when the transporter activities go from near zero in micro-gravity to quite high as the g-force rises to 2-g or higher levels achieved during parabolic flight or on a centrifuge on the ISS.
Task Progress:	Year one update on progress on three specific aims of proposal Aim 1. Use proteomic, immunological, and yeast-based activity assays to determine whether cation transporters that are known to be expressed at the peak of the trans-cell Ca2+ current are localized on the plasma membrane and are regulated post-translationally. Progress: Prior work has identified a specific calcium pump, CrACA1, as playing a major role in driving the trans-cell calcium current that is directed by gravity in fern spores cells. During this first year of the grant we have further characterizied antibodies to this calcium pump by using them in immunoprecipitation experiments to rapidly separate the pump from other proteins in extracts of fern spores. In these experiments the non-ionic detergent Brij-35 is used to solubilize membranes and allows immunoprecitation of integral membrane proteins as well as cytosolic proteins. We found that the immune serum recognizes at least two bands that are absent in experiments in which pre-immune serum is used. Additionally these same products retain reactivity with the CrACA1 antibody as demonstrated by western blot analysis of immunoprecipitation products. Biochemical identification of these bands is underway. Once recognition of the CrACA1 protein by the antibody has been verified, the antibody will be used to obtain sufficient quantity of the protein for evaluation of post-translational modifications to various stages in the course of gravity response. To do these experiments, novel methods were developed for protein isolation from spores that are in a fixed orientation in agar growth media. Dr. Mari Salmi, the research scientist working on this project, has been selected to attend the 2014 workshop on Plant Quantitative Proteomics at the University of Wisconsin in July 2014. She will get hands on training in mass spectrometric techniques, post-translational modification identification, and global proteome quantification. The knowledge and skills gained at this workshop will be applied to evaluation of the proteins involved in regulating the trans-cellular calcium current. Aim 2. Test the role of Ca2+-binding proteins expressed at the peak of the trans-cell Ca2+ current by suppressing their expression using transformation methods. Progress: Late during this first year of the grant a new, promising method for genetically transforming Ceratopteris spores was published. In preparation for using this new method we are currently creating new plasmid constructs that will allow us to stably suppress two specific genes in Ceratopteris that encode proteins implicated as important for the trans-cell Ca2_ current, annexin and CrACA1. Aim 3. Test the role of eATP in promoting the critical Ca2+-uptake step of the spore gravity response, using extracellular luciferase reporters to assay whether the gravity stimulus induces a top-to-bottom gradient of eATP, and by intracellular reporters to assay changes in [Ca2+]cyt. Progress: As described in our proposal we have established a collaboration with Dr. Eric McLamore at The University of Florida to use an ATP-specific microelectrode, to measure the [eATP] on the top and bottom of spores. In preliminary trials, the [eATP] at the bottom of the spore is over two-times higher then the concentration at the top. This exciting result could be evidence of a role for eATP in the Ca2+ uptake step, for an asymmetric gradient of eATP could direct the uptake preferentially along the bottom of the spore, a key step in initiating the trans-cell calcium current. We want to confirm the microelectrode results with an independent method. Because firefly luciferase enzymes give off light in proportion to the [ATP] surrounding them, they have been used to report the [eATP] concentration in and outside of cells. We have constructed a reporter comprised of a luciferase protein with a cellulose-binding domain, expressed it in E. coli, and purified it using a published protocol. Currently, we are optimizing the protocol for cellulose-binding and luminescent imaging using the NightOWL imaging system (Berthold Technologies). The reporter will be applied to spores and used to assay for a top-to-bottom gradient of eATP in response to gravity, which would be an independent test of the ATP-specific microelectrode results described above. We are also developing a method to visualize Ca2+ as it crosses the spore cell in the gravity-directed trans-cell current. Calcium Crimson (Life Technologies), a long-wavelength calcium indicator, is being tested for use as an intracellular calcium reporter in spores. The spores are mixed with the indicator and placed at different temperatures for varying

	periods of time based on the pollen loading protocol in Qu et al. (Qu et al., 2012). Spores have been screened for intracellular Calcium Crimson using an Axiovert 200 M Fluorescence Microscope (Zeiss), and initial experiments indicate a strong Ca2+ signal can be visualized. After this protocol is optimized, a confocal microscope will be used to assay changes in [Ca2+}cyt in response to uniform and unilateral ATP application. Other accomplishments: We have successfully re-established greenhouse procedures for growing Ceratopteris sporophytes and harvesting from them abundant spores that can be used for not only our experiments but also for experiments planned for SporeSat assays of the effects of different g-forces on the trans-cell calcium current. We have completed 5 manuscripts (two primary research articles published, plus another one submitted, and two reviews to be submitted in July) related to our project. Qu HY, Jiang XT, Shi ZB, Liu LM, Zhang SL (2012) Fast loading ester fluorescent Ca2+ and pH indicators into pollen of Pyrus pyrifolia. Journal of Plant Research 125:185--195
Bibliography Type:	Description: (Last Updated: 10/16/2023)
Abstracts for Journals and Proceedings	Cannon AE, Clark G, Roux SJ. "A Gradient of Extracellular Nucleotides Directs Polarization in Early Growth and Development of Ceratopteris Spores." Presented at the 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 77. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Articles in Peer-reviewed Journals	Roux S. "A start point for extracellular nucleotide signaling." Molecular Plant. 2014 Jun;7(6):937-8. Epub 2014 Mar 11. PMID: 24618880 ; http://dx.doi.org/10.1093/mp/ssu023 , Jun-2014
Articles in Peer-reviewed Journals	Bushart TJ, Cannon A, Clark G, Roux SJ. "Structure and function of CrACA1, the major PM-type Ca2+-ATPase, expressed at the peak of the gravity-directed trans-cell calcium current in spores of the fern Ceratopteris richardii." Plant Biology (Stuttg). 2014 Jan;16 Suppl 1:151-7. Epub 2013 Oct 29. http://dx.doi.org/10.1111/plb.12107 ; PubMed PMID: 24373013 , Jan-2014